EVER GUARDIAN

System Subsystems Description

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Subsystem 1 — Sensing

Sensor layer — physical and physiological data acquisition via I²C and UART.

MAX30102 — Heart Rate & SpO_2

MAX30102 — Heart Rate & SpO_2	
Interface	I²C — Address 0x57
Sampling Rate	100 Hz (configurable 50–3200 Hz)
ADC Resolution	18-bit
Supply Voltage	3.3V I/O
Current	~10 mA active
Alert Trigger	HR < 45 or > 130 BPM (30s) | SpO_2 < 90%

MPU6050 — Accelerometer + Gyroscope

MPU6050 — Accelerometer + Gyroscope (IMU)	
Interface	I²C — Address 0x68
Accel Range	±16g (fall detection)
Gyro Range	±2000°/s
ADC Resolution	16-bit
Current	~4 mA active
Fall Detection	Phase 1: |a|>2.5g | Phase 2: |a|<0.3g or gyro>200°/s | Phase 3: stillness

MAX30205 — Body Temperature

MAX30205 — Human Body Temperature Sensor	
Interface	I²C — Address 0x48–0x4B
Range	0°C to +50°C
Accuracy	±0.1°C between 37–39°C
Resolution	0.00390625°C (16-bit)
Current	~600 µA active
Alert Trigger	Temp < 35°C (hypothermia) | > 38°C (fever)

NEO-6M — GPS

NEO-6M — GPS Module	
Interface	UART — 9600 baud

Protocol	NMEA 0183 — GPRMC & GPGGA
Accuracy	~2.5 m CEP
Cold Start	< 27 seconds
Current	~45 mA
Update Rate	1 Hz — location included in every alert

Safety Button — SOS

Safety Button — Emergency Trigger	
Interface	GPIO34 — INPUT_PULLUP
Debounce	50 ms hardware debounce
Alert Type	CRITICAL — sent via Wi-Fi AND GSM simultaneously

Sensing Summary

Sensing Subsystem — All Sensors Summary	
Sensor	**Interface | Measurement | Current | Alert**
MAX30102	I²C 0x57 | PPG 100 Hz | ~10 mA | HR<45/>130, SpO_2<90%
MPU6050	I²C 0x68 | ±16g / ±2000°/s | ~4 mA | Fall 3-phase
MAX30205	I²C 0x48 | Temp ±0.1°C | ~600 µA | <35°C or >38°C
NEO-6M	UART 9600 | Lat/Lon NMEA | ~45 mA | Geofence breach
SOS Button	GPIO34 | Binary 0/3.3V | ~0 mA | Long press

Subsystem 2 — Processing

ESP32-WROOM-32D — dual-core FreeRTOS — real-time computation and alert decisions.

ESP32-WROOM-32D Specifications

ESP32-WROOM-32D — Microcontroller	
CPU	Dual-core Xtensa® LX6 — 240 MHz each
SRAM / Flash	520 KB SRAM | 4 MB Flash
Wi-Fi	802.11 b/g/n — 2.4 GHz — MQTT over TCP
GPIO	34 programmable pins
I²C	2 hardware controllers — 400 kHz
UART	3 ports — UART2 used for GPS
Current	~240 mA (Wi-Fi active) | ~0.3 mA (deep sleep)
OS	ESP-IDF with FreeRTOS v10

FreeRTOS Tasks

FreeRTOS Task Architecture	
Task | Core | Priority	**Responsibility**
sensorTask | Core 0 | High	Read all sensors via I²C and UART into ring buffer — 10 ms
fallDetectTask | Core 1 | Critical	3-phase fall algorithm on 5s IMU history window — 10 ms
vitalMonitorTask | Core 1 | High	Calculate HR, SpO_2 — check all vital thresholds — 1000 ms
alertTask | Core 1 | Critical	Build JSON payload — select Wi-Fi or GSM — send alert
wifiManagerTask | Core 0 | Medium	Monitor Wi-Fi — trigger GSM fallback if 3 MQTT failures
gsmTask | Core 0 | Medium	Handle SIM800L AT command sequences for GSM transmission

Signal Processing Pipeline

6-Step Processing Pipeline	
Step 1 — Acquisition	sensorTask reads raw 16-bit values at 100 Hz → circular buffer in SRAM
Step 2 — Filtering	Butterworth LPF on accelerometer | Moving average on PPG signal
Step 3 — Feature Extraction	Peak detection on IR PPG → IBI → HR | AC/DC ratio Red/IR → SpO_2
Step 4 — Fall Detection	3-phase state machine on filtered accel + gyro data
Step 5 — Threshold Check	Vital signs compared to thresholds in config.h
Step 6 — Alert Decision	alertTask builds JSON payload — selects Wi-Fi or GSM channel

Subsystem 3 — Communication

Dual-path Wi-Fi (primary) + GSM/SIM800L (fallback) — reliable alert delivery to guardian.

Wi-Fi Path — Primary Channel	
Standard	IEEE 802.11n — 2.4 GHz
Protocol	MQTT over TCP — port 1883
Backend	Firebase Realtime Database + Cloud Functions
Push Notification	Firebase Cloud Messaging (FCM) → Flutter app
Latency	< 100 ms end-to-end
Current	~80 mA (transmitting)

GSM Path — Fallback Channel (SIM800L)	
Module	SIM800L — Quad-band GSM/GPRS 850/900/1800/1900 MHz
Protocol	GPRS Class 10 — HTTP POST via AT commands
Endpoint	Firebase REST API
Interface	UART — 9600 baud
Supply	4.0V from MT3608 boost converter
Current	~200 mA idle | 2A peak TX burst
Latency	< 500 ms

Wi-Fi vs GSM Comparison	
Parameter	**Wi-Fi (Primary) | GSM / SIM800L (Fallback)**
Speed	72 Mbps (802.11n) | 85.6 kbps (GPRS Class 10)
Coverage	~50m indoor from router | National cellular network
Latency	< 100 ms | < 500 ms
Current	~80 mA | ~200 mA idle / 2A TX peak
Activation	Default when SSID available | 3 MQTT failures or CRITICAL alert

Subsystem 4 — Power

LiPo 1200 mAh — USB-C charging — dual-rail distribution (3.3V + 4.0V).

Power Chain Components	
LiPo Battery	3.7V nominal | 1200 mAh | 4.44 Wh | 55×35×5 mm
TP4056 Charger	USB-C 5V input | 1A charge | 4.2V CV | CC→CV→Trickle
DW01A BMS	Overcharge >4.28V | Over-discharge <2.9V | Overcurrent >3A | Short-circuit
AMS1117-3.3 LDO	3.3V fixed | 800 mA max | Powers ESP32 + all sensors
MT3608 Boost	Input 2.5–4.2V → Output 4.0V | 2A continuous | Powers SIM800L only

Current Consumption Budget			
Component	**3.3V Rail**	**4.0V Rail**	**Notes**
ESP32-WROOM-32D	~240 mA	—	Wi-Fi active + CPU full speed
MAX30102	~10 mA	—	LED on, 100 Hz sampling
MPU6050	~4 mA	—	Gyro + Accel active
MAX30205	~600 µA	—	Measurement every 5s
NEO-6M GPS	~45 mA	—	Active acquisition
SIM800L (idle)	—	~10 mA	GPRS registered, no TX
SIM800L (TX peak)	—	~2000 mA	100µF cap absorbs burst
3.3V Rail TOTAL	~321 mA	—	All sensors + ESP32 active

Battery Life Estimation			
Mode	**Current**	**Duration**	**Condition**
Full Active (Wi-Fi + GPS)	~321 mA	~3.7 h	All sensors active
Wi-Fi (no GPS)	~276 mA	~4.3 h	Indoor use
GSM Active	~350 mA avg	~3.4 h	All sensors + SIM800L
Light Sleep	~15 mA	~80 h	IMU only, Wi-Fi sleep
Deep Sleep (night)	~0.3 mA	~4000 h	RTC only
Target Daily Profile	~160 mA avg	~7.5 h	50% active / 50% sleep

Subsystem 5 — Wearable Enclosure

3D-printed wristwatch — PCB stack — 22mm silicone strap — target < 45g.

Mechanical Specifications	
Case Material	3D-printed PLA / PETG — lightweight, reprinta ble
Dimensions	~52 mm × 42 mm × 15 mm
Strap	22 mm standard silicone — hypoallergenic — replaceable
SOS Button	Right side — recessed ring guard (no accidental press)
Charging	USB-C — bottom edge
Target Weight	< 45 g with all components
IP Protection	IP53 — dust + splash resistant

PCB Design	
Dimensions	40 mm × 35 mm (target)
Layers	2-layer — components top / ground plane bottom
Tool	KiCad EDA
Fabrication	JLCPCB or local manufacturer
I²C Routing	Shared SDA/SCL with 4.7 kΩ pull-up to 3.3V

PCB Component Layer Stack	
Layer	**Components | Thickness**
Layer 1 (Top)	ESP32-WROOM-32D, MAX30102, MPU6050, MAX30205 | ~4 mm
Layer 2	SIM800L module | ~3 mm
Layer 3	NEO-6M GPS (antenna outward) | ~3 mm
Layer 4 (Bottom)	LiPo battery 1200 mAh | ~5 mm
TOTAL STACK	Including PCB + gaps | ~15–17 mm (target <12 mm after optimization)

Subsystems Integration Summary

All 5 Subsystems — Integration Overview	
Subsystem	**Key Components | Role | Connected To**
1 — Sensing	MAX30102, MPU6050, MAX30205, NEO-6M,safety button | Raw data acquisition | ESP32 via I²C/UART
2 — Processing	ESP32-WROOM-32D — FreeRTOS — 6 tasks | Detection + alert decisions | Communication subsystem
3 — Communication	ESP32 Wi-Fi + SIM800L GSM | Reliable alert delivery | Firebase → FCM → Guardian App
4 — Power	LiPo + TP4056 + DW01A + AMS1117 + MT3608 | 3.3V and 4.0V rails | All subsystems
5 — Enclosure	PCB 40×35mm + 3D case + 22mm strap | Physical packaging | User (physical interaction)